.NET MAUI Tutorial - Build your first multi-platform app in C#

Windows macOS

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Download and install

1. Download and install the .NET SDK. After the download completes, launch the download to begin the install.

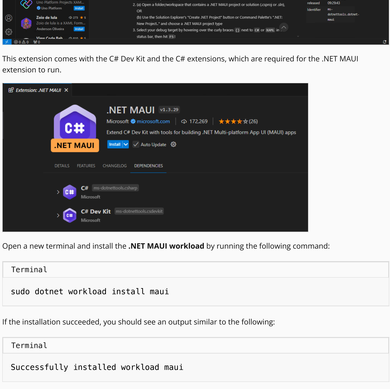
   Download .NET 9 SDK x64 (Intel) →
   Download .NET 9 SDK Arm64 (Apple Silicon)

   > If you're on a Mac with an Apple M1 or M2 chip, you need to install the Arm64 version of the SDK.

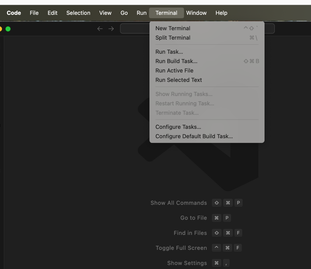
2. Download and install VS Code:

   Download VS Code →

3. Open VS Code and select the **Extensions button** in VS Code's Activity Bar to the left. Type **MAUI** in the search bar, select .NET MAUI, and then select the **Install** button on the .NET MAUI extension page.

   This extension comes with the C# Dev Kit and the C# extensions, which are required for the .NET MAUI extension to run.

4. Open a new terminal and install the **.NET MAUI workload** by running the following command:

   ```
   Terminal

   sudo dotnet workload install maui
   ```

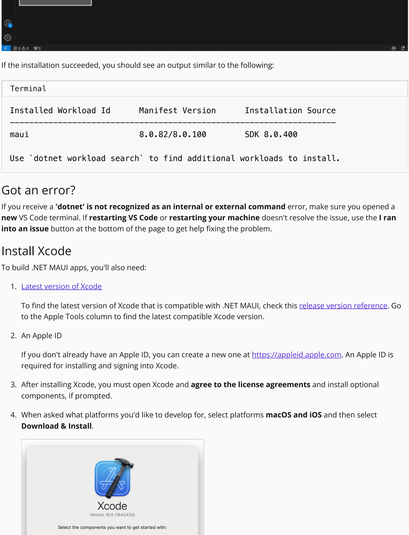
   If the installation succeeded, you should see an output similar to the following:

   ```
   Terminal

   Successfully installed workload maui
   ```

Check everything installed correctly

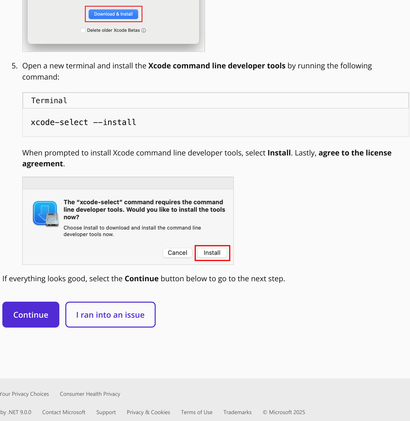
Once you've set up your environment, open a new VS Code terminal. In the toolbar, select **Terminal**, then **New Terminal**.



In your terminal, run the following command to check your installation.

```
Terminal

dotnet workload list
```



If the installation succeeded, you should see an output similar to the following:

```
Terminal

Installed Workload Id       Manifest Version      Installation Source
----------------------------------------------------------------------
maui                        8.0.82/8.0.100        SDK 8.0.400

Use `dotnet workload search` to find additional workloads to install.
```

Got an error?

If you receive a **'dotnet' is not recognized as an internal or external command** error, make sure you opened a **new** VS Code terminal. If **restarting VS Code** or **restarting your machine** doesn't resolve the issue, use the **I ran into an issue** button at the bottom of the page to get help fixing the problem.

Install Xcode

To build .NET MAUI apps, you'll also need:

1. Latest version of Xcode

   To find the latest version of Xcode that is compatible with .NET MAUI, check this release version reference. Go to the Apple Tools column to find the latest compatible Xcode version.

2. An Apple ID

   If you don't already have an Apple ID, you can create a new one at https://appleid.apple.com. An Apple ID is required for installing and signing into Xcode.

3. After installing Xcode, you must open Xcode and **agree to the license agreements** and install optional components, if prompted.

4. When asked what platforms you'd like to develop for, select platforms **macOS and iOS** and then select **Download & Install**.

5. Open a new terminal and install the **Xcode command line developer tools** by running the following command:

   ```
   Terminal

   xcode-select --install
   ```

   When prompted to install Xcode command line developer tools, select **Install**. Lastly, **agree to the license agreement**.

If everything looks good, select the **Continue** button below to go to the next step.

Continue | I ran into an issue